

Foundation Programme
Universiti Teknologi Malaysia
Session 2021/2022 Semester 3

Test 2 (20%)

Time: 90 minutes

FSPM 0024 Calculus

Answer all questions. Show all your work clearly. Any similarity with answers given by apps/chegg or any websites will be punished accordingly. List of formula is available on the last page.

1. Given $\int_0^5 g(x) dx = 2$, $\int_0^3 g(x) dx = -4$, $\int_3^5 [5g(x) - 4h(x)] dx = 50$.

i. Find $\int_3^5 h(x) dx$.

(3 marks)

ii. Hence, evaluate $\int_3^5 \left[\frac{3}{2}x^3 - 11h(x) \right] dx$.

(3 marks)

2. By taking $u^2 = 3x + 1$, solve the integral

$$\int \frac{2x + 4}{\sqrt{3x + 1}} dx.$$

(5 marks)

3. Evaluate $\int e^{2x} \sin 4x dx$ by using tabular method.

(4 marks)

4. By using the substitution $u = e^x$, show that

$$\int_{-1}^1 \frac{e^x}{e^x + 1} dx = 1.$$

(5 marks)

$$1) \text{ ci) } 5 \int_3^5 g(x) - 4 \int_3^5 h(x) = 50$$

$$5 \left(\int_0^5 g(x) + \left(- \int_3^0 g(x) \right) \right) - 4 \int_3^5 h(x) = 50$$

$$5(2 - (-4)) - 4 \int_3^5 h(x) = 50$$

$$\int_3^5 h(x) dx = \frac{30-50}{4} \\ = -5$$

$$\begin{aligned} \text{cii) } \frac{3}{2} \int_3^5 x^3 - 11 \int_3^5 h(x) &= \frac{3}{2} \left[\frac{x^4}{4} \right]_3^5 - 11(-5) \\ &= \frac{3}{2} \left(\frac{625}{4} - \frac{81}{4} \right) + 55 \\ &= \frac{3}{2} (136) + 55 \\ &= 259 \end{aligned}$$

$$\begin{aligned} 2) \quad u^2 &= 3x+1, \quad \int \frac{2x+4}{\sqrt{3x+1}} dx = \int \frac{2x+4}{\sqrt{u^2}} \cdot \frac{2x}{3} \cdot du \\ \frac{d}{dx}(u^2) &= 3 &= \frac{2}{3} \int 2 \left(\frac{u^2-1}{3} \right) + 4 du \\ 2u \frac{du}{dx} &= 3 &= \frac{2}{3} \int \frac{2u^2-2}{3} + 4 du \\ dx &= \frac{2u}{3} \cdot du &= \frac{2}{3} \int \frac{2u^2-2}{3} + \frac{12}{3} du \\ x &= \frac{u^2-1}{3} &= \frac{2}{9} \int 2u^2 + 10 du \\ u &= \sqrt{3x+1} &= \frac{2}{9} \left(\frac{2u^3}{3} + 10u \right) + c \\ & &= \frac{4(\sqrt{3x+1})^3}{27} + \frac{20\sqrt{3x+1}}{9} + c \end{aligned}$$

$$3) \int e^{2x} \sin 4x dx = e^{2x} \left(\frac{-\cos 4x}{4} \right) - 2e^{2x} \left(\frac{-\sin 4x}{16} \right) + \int 4e^{2x} \left(\frac{-\sin 4x}{16} \right) dx$$

sign differentiate u integrate v'

$$+ \quad e^{2x} \quad \rightarrow \quad \sin 4x$$

$$- \quad 2e^{2x} \quad \rightarrow \quad \frac{-\cos 4x}{4}$$

$$+ \quad 4e^{2x} \quad \rightarrow \quad \frac{-\sin 4x}{16}$$

$$\int e^{2x} \sin 4x dx = \frac{-e^{2x} \cos 4x}{4} + \frac{e^{2x} \sin 4x}{8} - \frac{1}{4} \int e^{2x} \sin 4x dx$$

$$\frac{5}{4} \int e^{2x} \sin 4x dx = -\frac{1}{4} (e^{2x} \cos 4x) + \frac{1}{8} (e^{2x} \sin 4x)$$

$$\int e^{2x} \sin 4x dx = -\frac{1}{5} (e^{2x} \cos 4x) + \frac{1}{10} (e^{2x} \sin 4x)$$

$$4) \quad u = e^x \quad \int_{-1}^1 \frac{e^x}{e^x + 1} dx = \int_{-1}^1 \frac{e^x}{e^x + 1} \cdot \frac{du}{e^x} = \int_{-1}^1 \frac{1}{u+1} du = [\ln u + 1]_{-1}^1 = [\ln(e^x + 1)]_{-1}^1 = 1.313 - 0.313 = 1$$

$$\frac{du}{dx} = e^x$$

$$dx = \frac{du}{e^x}$$

$$5) \quad f(x) = \frac{12x^2 + x + 3}{(6x+1)(2x^2+1)} = \frac{A}{6x+1} + \frac{Bx+C}{2x^2+1} = \frac{3}{6x+1} + \frac{x}{2x^2+1}$$

$$A(2x^2+1) + Bx + C(6x+1) = 12x^2 + x + 3$$

$$(2A+6B)x^2 + (B+6C)x + (A+C) = 12x^2 + x + 3$$

$$A+C=3 \quad B+6(3-A)=1 \quad 2A+6(-17+6A)=12$$

$$C=3-A \quad B+18-6A=1 \quad 2A-102+36A=12$$

$$C=0 \quad B-6A=-17 \quad 38A=114$$

$$A=3$$

$$B=-17+6A$$

$$B=-17+6(3)$$

$$B=1$$

$$\int_0^2 f(x) dx = \int_0^2 \frac{3}{6x+1} dx + \int_0^2 \frac{x}{2x^2+1} dx$$

$$u = 2x^2 + 1, \quad \frac{du}{dx} = 4x, \quad dx = \frac{du}{4x}$$

$$= 3[\ln(3x+1)]_0^2 + \int_0^2 \frac{x}{2x^2+1} \cdot \frac{du}{4x}$$

$$= 3\ln(7) + 0 + \frac{1}{4} \int_0^2 \frac{1}{u} du$$

$$= 3\ln(7) + \frac{1}{4} [\ln 2x^2 + 1]_0^2$$

$$= 3\ln 7 + \frac{1}{4} \ln(7)$$

$$= 3\frac{1}{4} \ln 7$$

$$6) \quad (i) \quad -x^2 + 20 = x^3 + 8 \quad p(2, 16) \quad x^3 + 8 = 0 \quad q(-2, 0)$$

$$x^3 + x^2 - 12 = 0$$

$$x^3 = -8$$

$$(x-2)(x^2+3x+6) = 0$$

$$x = -2$$

$$x-2=0, \quad x=2$$

$$x^2+3x+6=0, \quad x \notin \mathbb{R}$$

$$(ii) \quad \text{Area A} = \left| \int_{-2}^0 x^3 + 8 dx \right|$$

$$= \left| \left[\frac{x^4}{4} + 8x \right]_{-2}^0 \right|$$

$$= \left| (0-0) - (4-16) \right|$$

$$= 12 \text{ unit}^2$$

$$\text{Area B} = \int_0^2 -x^2 + 20 - x^3 - 8$$

$$= \left[-\frac{x^3}{3} + 20x - \frac{x^4}{4} - 8x \right]_0^2$$

$$= 17.33 \text{ unit}^2$$

$$\text{Total} : 29.33 \text{ unit}^2$$

5. Given

$$f(x) = \frac{12x^2 + x + 3}{(6x + 1)(2x^2 + 1)}.$$

i. Find the partial fraction of $f(x)$.

(3 marks)

ii. Evaluate $\int_0^2 f(x) dx$. Give your answer in terms of natural logarithm.

(5 marks)

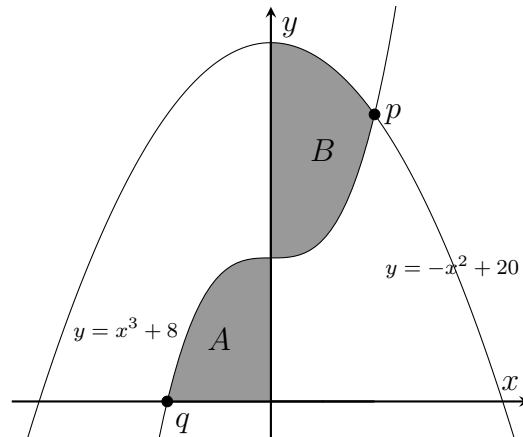


Figure 1: Figure for Question 6

6. Figure 1 shows the regions A and B that are bounded by the curves $y = x^3 + 8$, $y = -x^2 + 20$, x -axis and y -axis. Point p is the intersection point between the two curves while point q is located on the x -axis.

i. State the x -coordinates for p and q .

(2 marks)

ii. Find the total area of region A and B .

(5 marks)

iii. Find the volume of solid revolution of region A formed by revolving the region 360° about the y -axis.

$$\text{At } x=0, y=8 \quad x = \sqrt[3]{y-8}$$

(5 marks)

$$\begin{aligned} \text{Volume } A &= \pi \int_0^8 (\sqrt[3]{y-8})^2 dy \\ &= \pi \int_0^8 (y-8)^{\frac{2}{3}} dy \\ &= \pi \left[\frac{3}{5} (y-8)^{\frac{5}{3}} \right]_0^8 \\ &= \pi (0 - (-19.2)) \\ &= 60.32 \text{ unit}^2 \end{aligned}$$

LIST OF FORMULA

Trigonometric	
• $\cos^2 x + \sin^2 x = 1$ • $1 + \tan^2 x = \sec^2 x$ • $\cot^2 x + 1 = \operatorname{cosec}^2 x$ • $\sin 2x = 2 \sin x \cos x$ • $\cos 2x = \cos^2 x - \sin^2 x$ $= 2 \cos^2 x - 1$ $= 1 - 2 \sin^2 x$ • $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$	• $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$ • $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$ • $\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$ • $2 \sin x \cos y = \sin(x + y) + \sin(x - y)$ • $2 \sin x \sin y = -\cos(x + y) + \cos(x - y)$ • $2 \cos x \cos y = \cos(x + y) + \cos(x - y)$
Differentiations	Integrations
$\frac{d}{dx}[k] = 0, k \text{ constant}$	$\int k dx = kx + C, k \text{ constant}$
$\frac{d}{dx}[x^n] = nx^{n-1}$	$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
$\frac{d}{dx}[e^x] = e^x$	$\int e^x dx = e^x + C$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\int \frac{dx}{x} = \ln x + C$
$\frac{d}{dx}[\cos x] = -\sin x$	$\int \sin x dx = -\cos x + C$
$\frac{d}{dx}[\sin x] = \cos x$	$\int \cos x dx = \sin x + C$
$\frac{d}{dx}[\tan x] = \sec^2 x$	$\int \sec^2 x dx = \tan x + C$
$\frac{d}{dx}[\cot x] = -\operatorname{cosec}^2 x$	$\int \operatorname{cosec}^2 x dx = -\cot x + C$
$\frac{d}{dx}[\sec x] = \sec x \tan x$	$\int \sec x \tan x dx = \sec x + C$
$\frac{d}{dx}[\operatorname{cosec} x] = -\operatorname{cosec} x \cot x$	$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$